

SANDEEP KUMAR

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Summary

M.Tech in Communication Engineering from IIT Bombay with strong expertise in C/C++ embedded software development for Modem-RF Wireless systems. Hands-on experience in designing, implementing, and optimizing real-time 5G NR and LTE transceiver algorithms using GNU Radio and USRP B210, achieving 5 dB SNR improvement in MIMO-OFDM systems and 99.4% optimal throughput in mmWave beam selection. Deep knowledge of digital communications, RF theory, 3GPP standards, physical layer (PHY), OFDM, MIMO, and radio architecture. Proven ability to develop and validate Modem-RF drivers and algorithms in cross-functional environments. Seeking to contribute to Qualcomm's cutting-edge Modem-RF Transceiver Software team as a Software Engineer.

Education

- M.Tech in Communication Engineering, Indian Institute of Technology, Bombay July 2023 – June 2025
- B.Tech in Electronics Engineering, KNIT Sultanpur July 2015 – May 2019

Technical Skills

Technical: Python, C, C++, RLC, TCP/IP, Deep Neural Network, Model Training and Optimization, SQL, Pandas, NumPy

Software & Tools: GNU Radio, Gnuplot, Matplotlib, HFSS, Machine Learning, MATLAB, LaTeX, Excel

Wireless & RF: 5G NR, LTE, OFDM, MIMO, Modem-RF, RF Transceiver, Beamforming, mmWave, Physical Layer, Digital Communications, Communication Theory, Radio Architecture, 3GPP Standards

Experience

Project Staff Engineer | IIT Bombay August 2025 – Present

- Worked with 3GPP (Release 15–18) and 3GPP2 specifications for Wireless Communication Systems, including 5G NR, mmWave, OFDM and MIMO technologies
- Developed a two-stage optimization algorithm for LEO satellite constellations that reduced active satellite count by 41% and minimized total power consumption while maintaining QoS

Projects

Deep Learning for Joint User and Beam Selection pair in mmWave Networks January 2024 – June 2025

- Implemented an MCMC algorithm for Joint UE-Beam selection, achieving near-optimal throughput
- Developed and trained a Deep Neural Network to replace the MCMC algorithm, resulting in near-optimal throughput while maximizing SNR, weighted sum rate, and network efficiency
- Explored applications in 5G NR and beyond 5G (6G) systems for beam management and resource allocation

5G NR OFDM System with Cyclic Prefix November 2023

- Developed end-to-end OFDM transceiver supporting QPSK, 16QAM, 64QAM
- Modeled multipath channels and performed BER analysis

SDR Implementation of OFDM Systems January 2024

- Implemented OFDM transmitter and receiver in GNU Radio
- Generated OFDM signals, transmitted and received using USRP, recovered data at receiver, and visualized power spectrum and constellation diagrams
- Explored real-world challenges of OFDM including time/frequency synchronization, multipath fading, and channel estimation, validating theory with SDR-based implementation

5G NR Physical Layer (PHY) Modeling and Analysis June 2024

- Benchmarked 5G uplink waveforms, demonstrating a 9 dB PAPR reduction with DFT-s-OFDM to enhance power amplifier efficiency and extend battery life for mobile devices
- Modeled 5G's flexible numerology, quantifying an 87% reduction in slot latency using 120 kHz subcarrier spacing to enable ultra-reliable low-latency communication (URLLC)
- Engineered a mini-slot scheduling simulation, cutting transmission delay by 85% compared to slot-based scheduling to facilitate immediate data transmission for mission-critical traffic

Single-Ended Cascode Common-Source Low Noise Amplifier January 2025 – May 2025

- Designed and Implemented achieving NF < 2 dB, Gain > 15 dB, and 50 Ω matching at 2.3–2.4 GHz with supply of 1.2 Volts.
- Performed stability, S-parameter, pre-layout & post-layout simulations ensuring performance under practical conditions
- Documented methodology, challenges, and final results in a detailed report, showcasing practical RFIC design flow from schematic to layout

Relevant Courses

- Wireless and Mobile Communications
- Statistical Signal Analysis
- Digital Message Transmission
- Radiating System
- DSP and Its Applications
- Computational Techniques in EE
- Communications Networks
- RF Communication Engineering
- R&D